DILLIVRY

Project Waypoint

Target-state event-driven architecture for real-time, multi-vehicle delivery



ARCHITECTURE REVIEW BOARD // RELEASE 1.0 // 26 SEP 2025

Designed by Cynthia Agwaraonye

// PLATFORM AT A GLANCE · FY2024

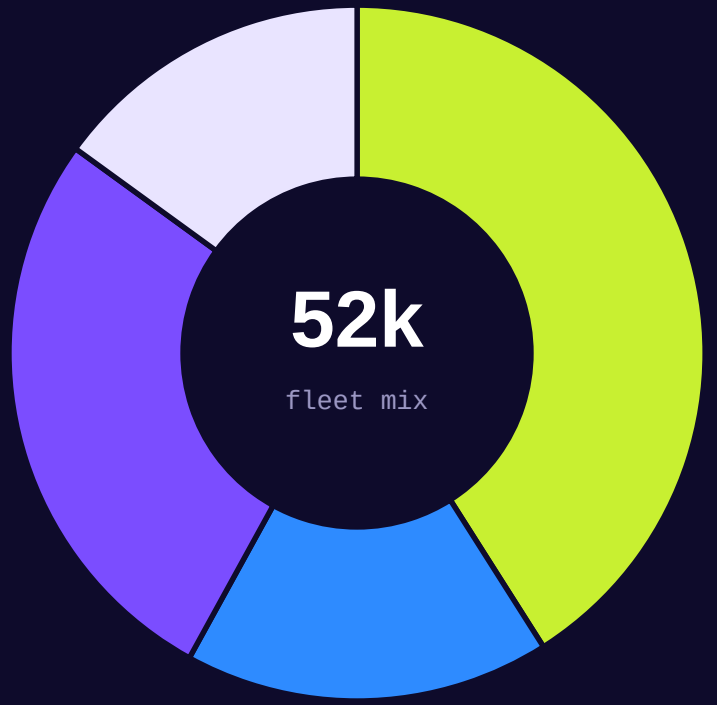
Four vehicle classes, one dispatch engine

38M
deliveries a year

52,000
active couriers

14
UK cities

1,250
orders / min at peak



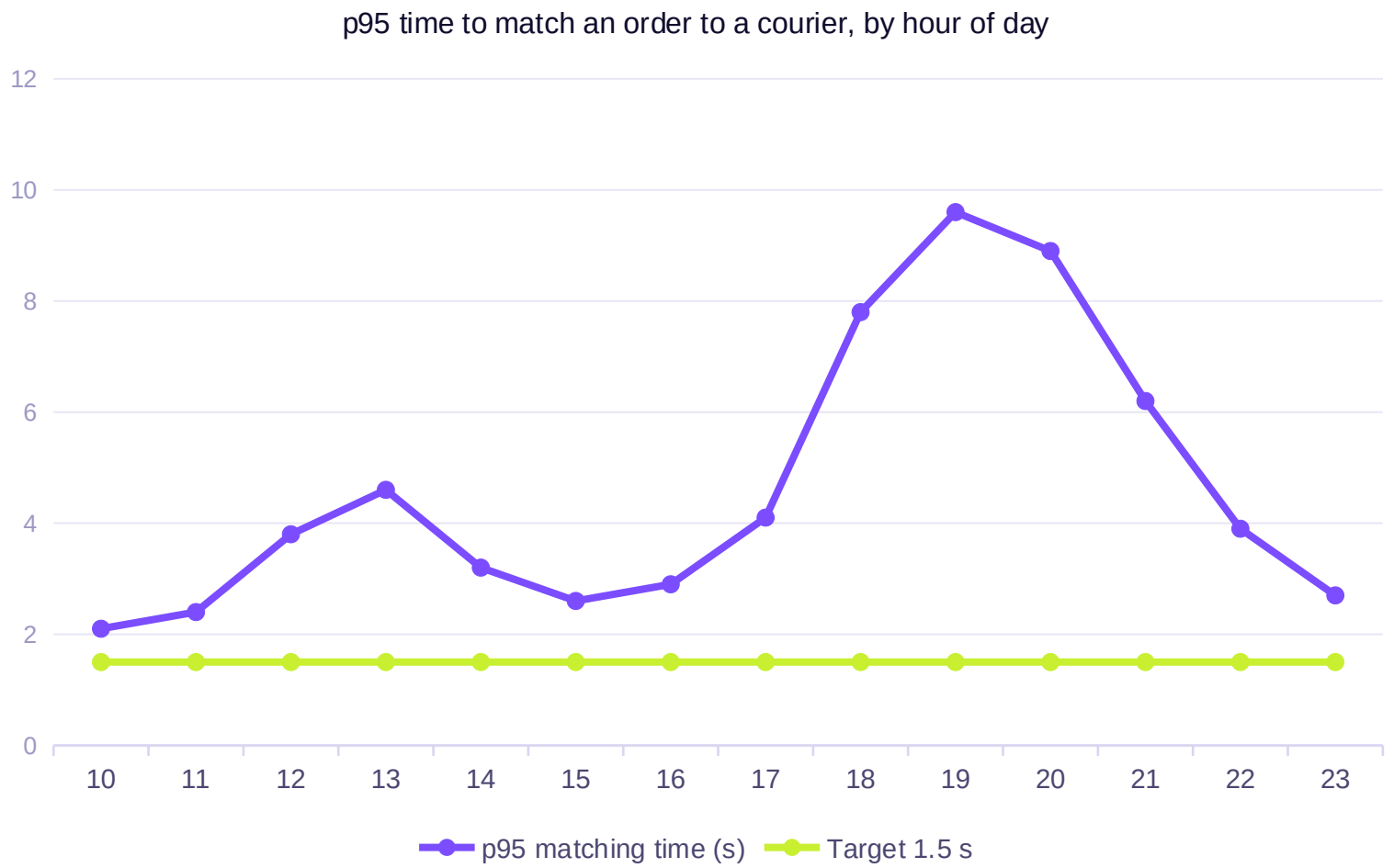
Bikes & e-bikes — ≤ 3 km · ≤ 8 kg — 41%

Motorbikes & scooters — ≤ 10 km · ≤ 15 kg — 17%

Cars — ≤ 25 km · ≤ 50 kg — 27%

Vans — bulky & multi-drop — 15%

Dispatch matching slows to 9.6 s at the evening peak



60 s

courier location polling interval

11 min

mean ETA error at peak

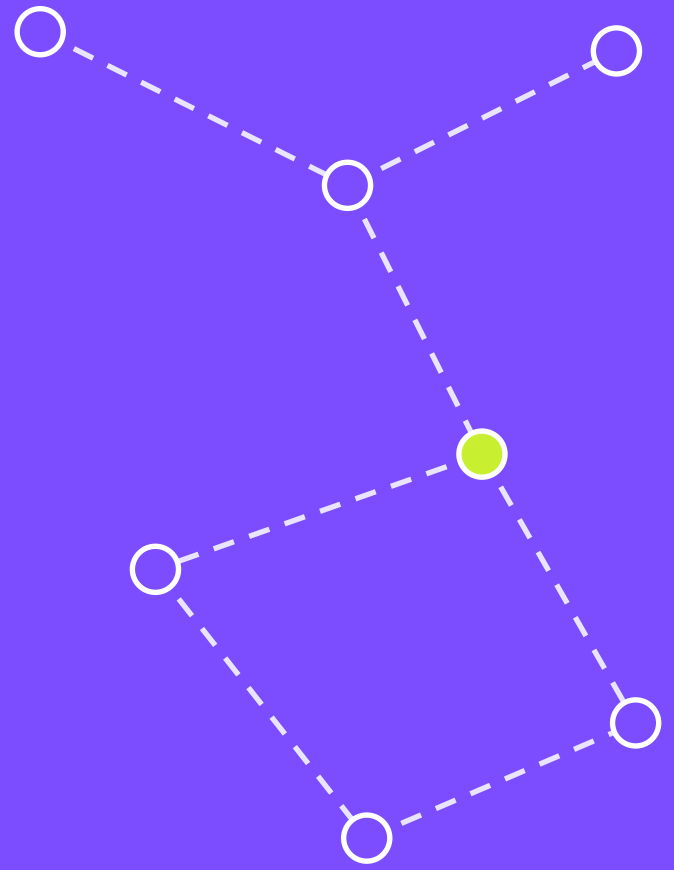
3

peak-time outages in Q4 2024

01

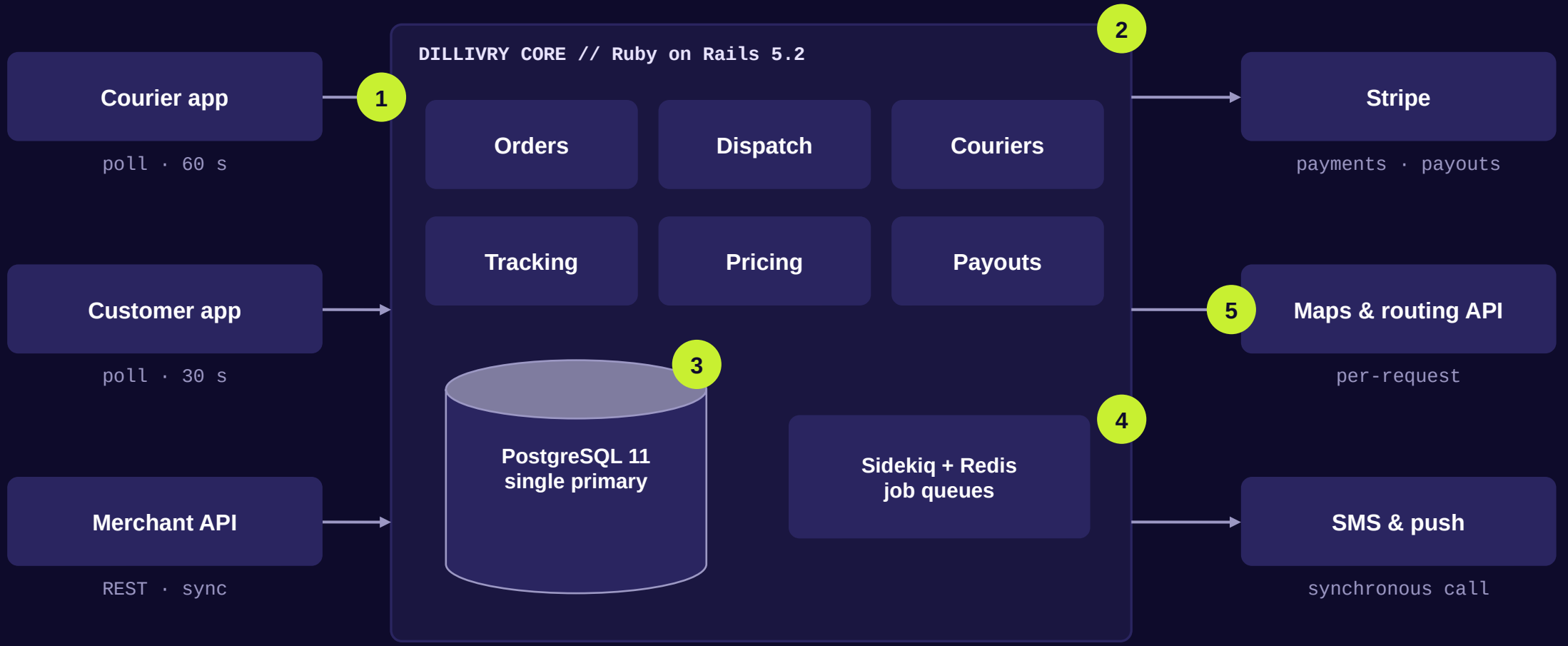
Current state

A 2016 monolith built for one city and one vehicle type



// CURRENT-STATE ARCHITECTURE

One Rails monolith, one database, polling everywhere



1 Polling makes locations up to 60 s stale

2 One deploy for all: 2-week release train

3 Single primary DB is the peak bottleneck

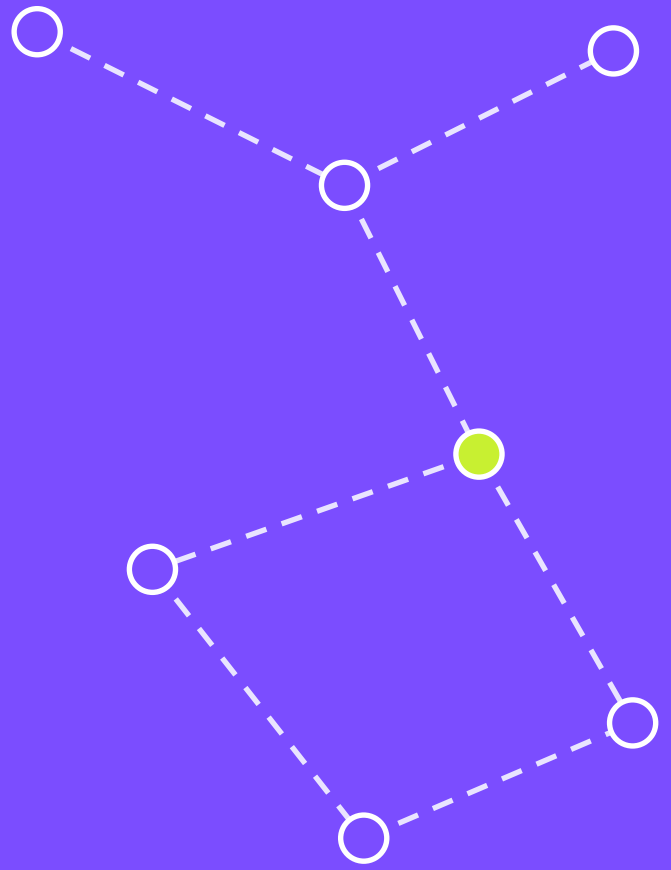
4 Queues back up at peak (34k jobs)

5 Routing calls block dispatch threads

02

Target architecture

Event-driven services built for the peak



// ARCHITECTURE PRINCIPLES

Six principles behind every design decision



P1

Events are the source of truth

Immutable, versioned domain events



P2

Own your data

No shared tables, no cross-service joins



P3

Design for the Friday peak

Autoscale on consumer lag, 3× headroom



P4

Contracts before code

OpenAPI 3.1 + Avro reviewed first



P5

Observable end to end

OpenTelemetry from rider to customer



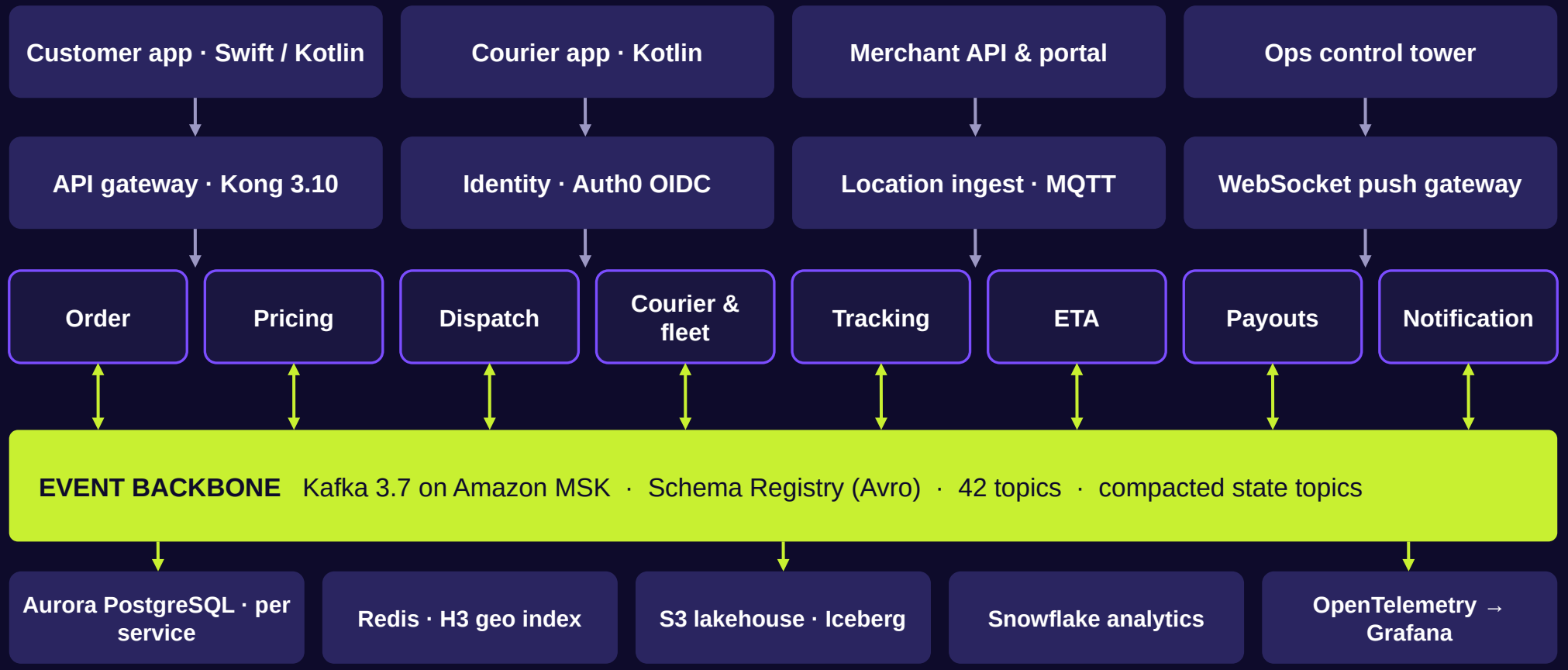
P6

Strangle, don't rewrite

Replace the monolith one capability at a time

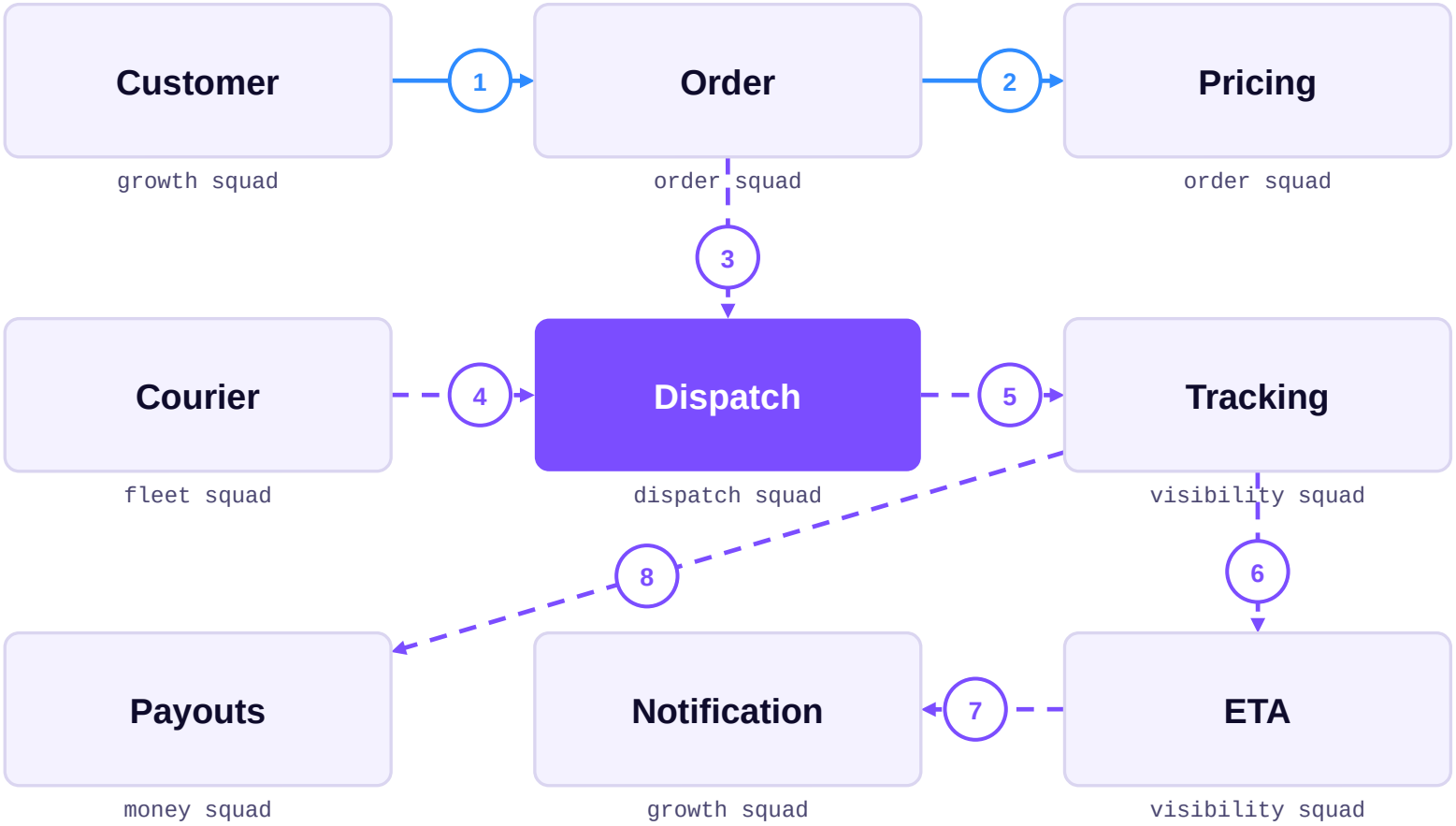
// TARGET-STATE LOGICAL ARCHITECTURE

Eight domain services on one Kafka event backbone



Stripe, maps and SMS providers connect through the edge only. No point-to-point calls into service databases.

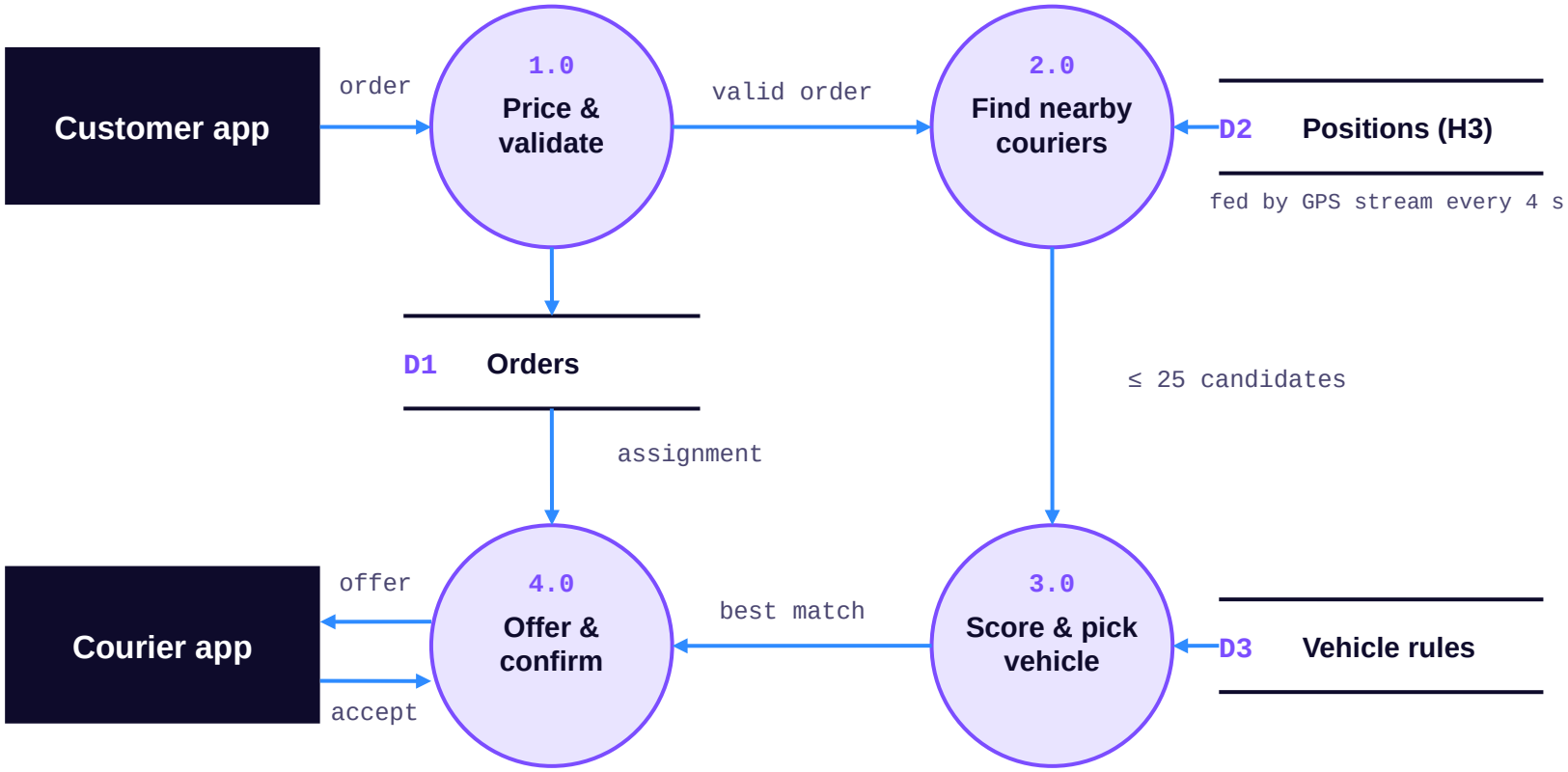
Nine bounded contexts, linked mostly by events



INTERACTIONS

1. **REST · place order**
   Customer → Order
2. **gRPC · quote**
   Order → Pricing
3. **`OrderPlaced`**
   Order → Dispatch
4. **`CourierAvailable`**
   Courier → Dispatch
5. **`JobAssigned`**
   Dispatch → Tracking
6. **`PositionUpdated`**
   Tracking → ETA
7. **`EtaRevised`**
   ETA → Notification
8. **`DeliveryCompleted`**
   Tracking → Payouts

Every order gets the right vehicle within 1.5 s



VEHICLE RULES (D3)

Bike / e-bike
≤ 3 km · ≤ 8 kg

Motorbike
≤ 10 km · ≤ 15 kg

Car
≤ 25 km · ≤ 50 kg

Van
bulky · multi-drop · ≤ 400 kg

9.6 s → 1.5 s
p95 match time, Friday peak

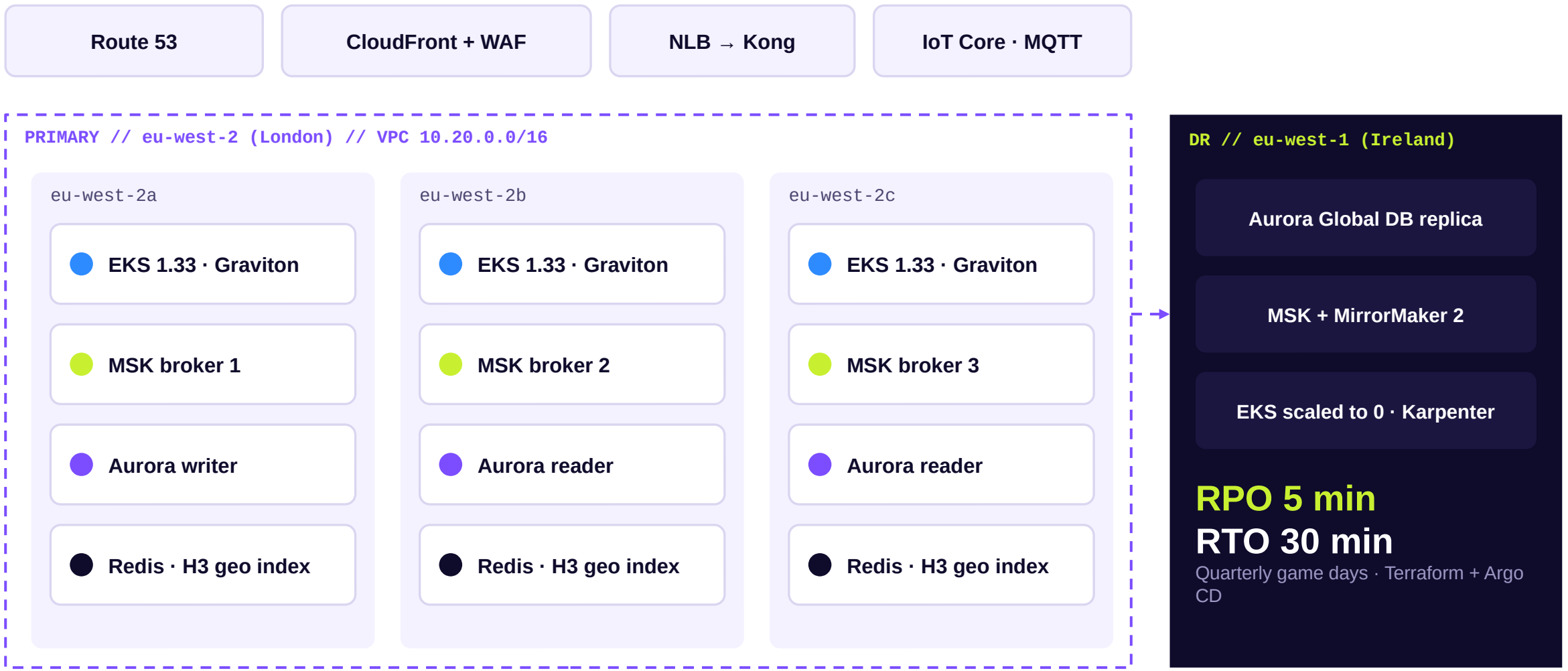
// EVENT FLOW · UML SEQUENCE

From courier GPS ping to customer ETA in 3 seconds



// CLOUD DEPLOYMENT TOPOLOGY

Three AZs in London, pilot-light recovery in Ireland



// NON-FUNCTIONAL REQUIREMENTS

Eight measurable quality targets

NFR-01

Availability

99.95%

order & tracking APIs, monthly

NFR-02

Location → customer

≤ 3 s

p95, GPS ping to app

NFR-03

Dispatch match

≤ 1.5 s

p95 at Friday peak

NFR-04

Throughput

25k/s

sustained events

NFR-05

Peak orders

3,750/min

3× Q4 2024 peak

NFR-06

Recovery

5 / 30 min

RPO / RTO, region loss

NFR-07

Security

TLS 1.3

PCI scope kept at Stripe

NFR-08

Interoperability

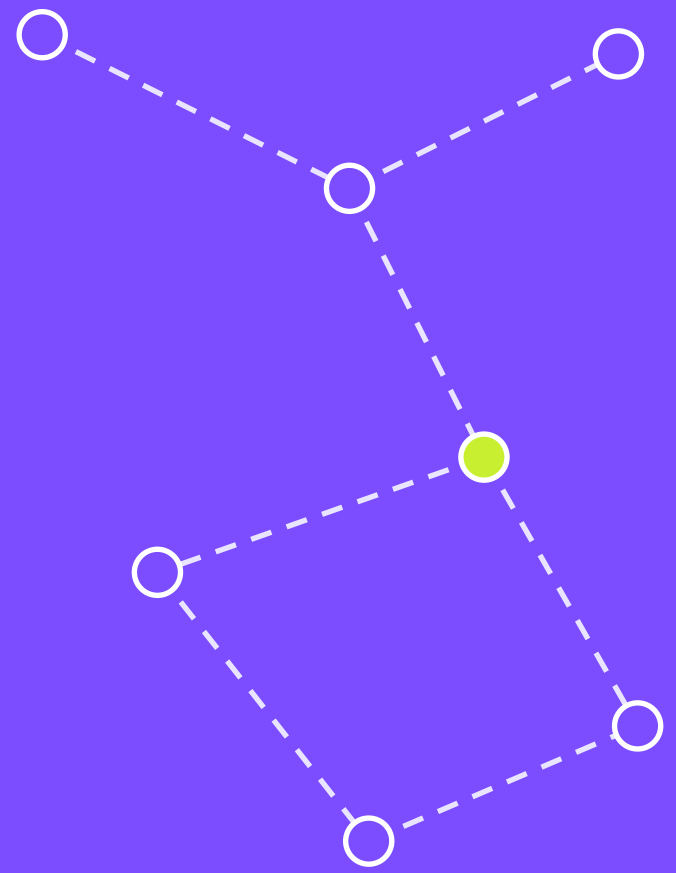
BACKWARD

Avro schema compatibility

03

Delivery plan

Migration, capacity, risk and decisions



// MIGRATION STRATEGY · STRANGLER FIG

Traffic moves off the monolith in four phases

PHASE 0

Oct–Dec 2025

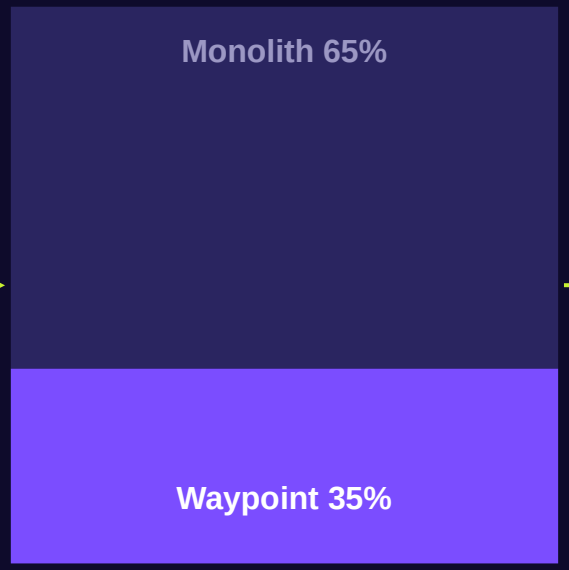


Foundations

MSK · EKS · CDC from Postgres

PHASE 1

Jan–Jun 2026

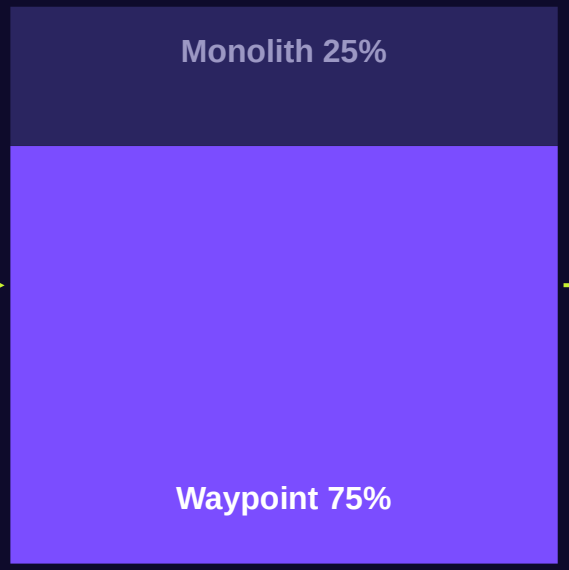


Live tracking

Tracking · ETA · Notification

PHASE 2

Jul–Dec 2026

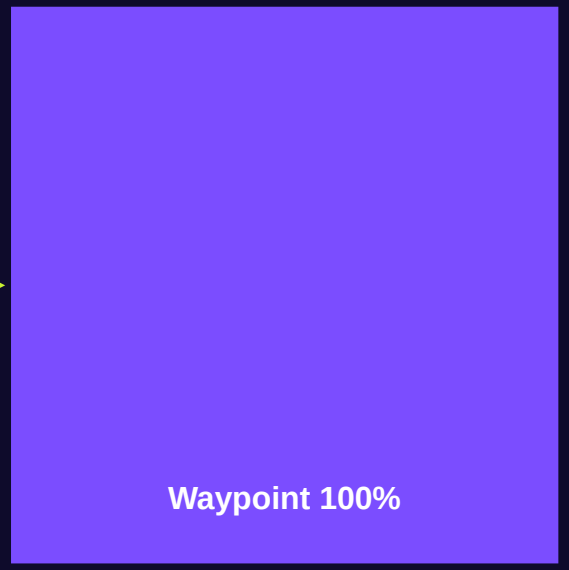


Dispatch

Dispatch · Courier · Order

PHASE 3

Jan–Sep 2027



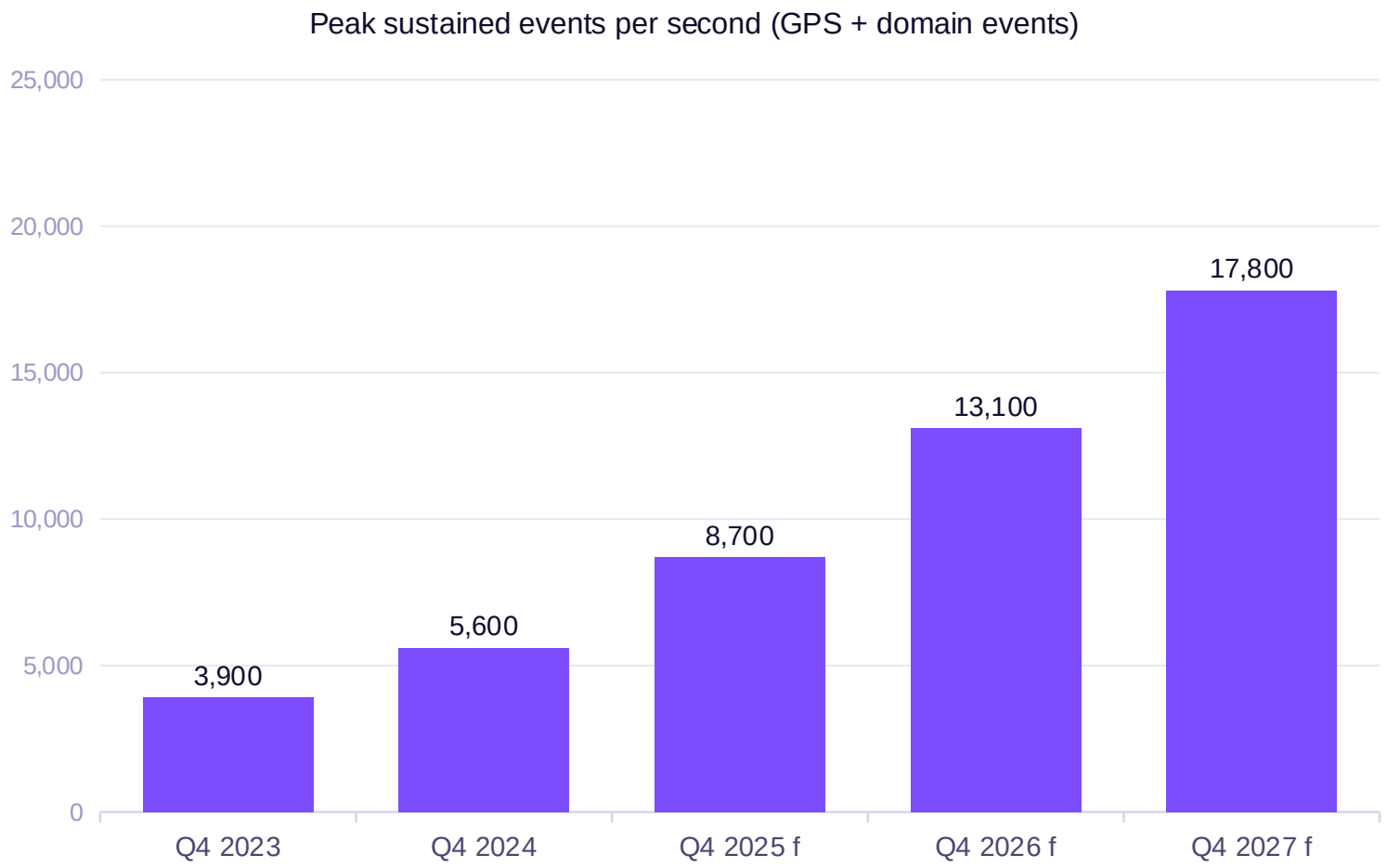
Retire

Pricing · Payouts · monolith off

Share of order volume per platform at phase exit. Kong routes each capability; Debezium CDC keeps both sides in sync.

// CAPACITY PLANNING

Peak load triples by 2027, still inside the ceiling



25,000

events/s design ceiling

1.4×

headroom over 2027 forecast

4 s

GPS interval, down from 60 s

// RISK REGISTER · SCORE = LIKELIHOOD × IMPACT (1–25)

Peak-season overlap is the top risk

RISK	SCORE	MITIGATION
Peak season overlaps Phase 0 Programme lead	20	Change freeze 17 Nov – 5 Jan; Phase 0 non-production only
Dual-write drift, monolith ↔ services Data architect	15	Debezium CDC + nightly reconciliation; alert at 0.01%
Courier app adoption of MQTT build Mobile lead	12	Staged rollout by city; 60 s polling fallback kept for 2 releases
Kafka skills gap in squads Eng. manager	12	Training for 26 engineers; platform-team pairing on-call
Cloud cost overrun FinOps lead	8	FinOps budgets per namespace; Graviton + Savings Plans

// ARCHITECTURE DECISION RECORDS

Six decisions for the board to ratify

ADR-001 — Accepted

Kafka (MSK) over RabbitMQ & Kinesis

replay · compaction · ecosystem

ADR-002 — Accepted

Avro + Schema Registry for events

typed, enforced compatibility

ADR-003 — Accepted

Database per service on Aurora

removes shared-schema coupling

ADR-004 — Accepted

H3 hexagonal index for courier search

O(1) neighbour lookup at city scale

ADR-005 — Proposed

MQTT for courier location

battery-friendly 4 s updates

ADR-006 — Proposed

Pilot-light DR in Ireland

RTO 30 min at 18% of active-active cost

// NEXT STEPS

Ratify Waypoint and start Phase 0 on 6 October

29 SEP
Board ratifies ADR-001 – 004

06 OCT
MSK + EKS in non-production

17 NOV
Peak-season change freeze

12 JAN
Phase 1 build: live tracking

Dillivry · Project Waypoint target-state architecture · Completed 26 September 2025

Designed by Cynthia Agwaraonye

Fictional client and illustrative data, prepared as a portfolio case study.